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Studierichting: Informatica
Oefeningenreeks 4A nr. 57.2

$$\int \frac{dx}{4\cos^2(x) + 9\sin^2(x)}$$

$$\sin^2(x) = \frac{1-\cos(2x)}{2}$$
$$\cos^2(x) = \frac{1+\cos(2x)}{2}$$

$$\int \frac{2dx}{4(1+\cos(2x)) + 9(1-\cos(2x))}$$
$$= \int \frac{2dx}{13 - 5\cos(2x)}$$

$$t = 2x$$
$$dt = 2dx$$

$$\int \frac{dt}{13 - 5\cos(t)}$$

$$u = \tan\left(\frac{t}{2}\right)$$
$$dt = \frac{2du}{1+u^2}$$
$$\cos(t) = \frac{1-u^2}{1+u^2}$$

$$\int \frac{\frac{2du}{1+u^2}}{13 - 5\left(\frac{1-u^2}{1+u^2}\right)}$$
$$= \int \frac{2du}{8 + 18u^2}$$
$$= \int \frac{du}{4 + 9u^2}$$

$$k = 3u$$
$$dk = \frac{1}{3}dk$$

$$
\begin{aligned}
&\frac{1}{3}\int \frac{du}{4+k^2}\\
&= \frac{1}{6}\,\mathrm{Bgtan}\left(\frac{k}{2}\right)+C\\
&= \frac{1}{6}\,\mathrm{Bgtan}\left(\frac{3u}{2}\right)+C\\
&= \frac{1}{6}\,\mathrm{Bgtan}\left(\frac{3\tan\left(\frac{t}{2}\right)}{2}\right)+C\\
&= \frac{1}{6}\,\mathrm{Bgtan}\left(\frac{3\tan\left(\frac{2x}{2}\right)}{2}\right)+C\\
&= \frac{1}{6}\,\mathrm{Bgtan}\left(\frac{3\tan(x)}{2}\right)+C
\end{aligned}
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References